

ADVANCED

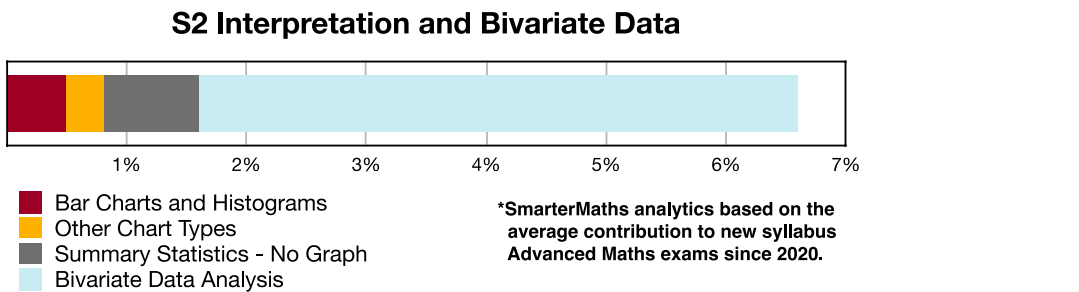
Stage 6

Statistics (Adv), S2 Interpretation and Bivariate Data (Adv)

Summary Statistics - Box Plots (Y12)

Teacher: Manjeet Kaur

Exam Equivalent Time: 55.5 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- S2 Interpretation and Bivariate Data* has contributed an average of 6.5% per new syllabus Advanced exam since it began in 2020.
- We have split the area into six sub-topics for analysis purposes which are: 1-*Classifying Data* (0%), 2-*Bar Charts and Histograms* (0.5%), 3-*Other Chart Types* (0.3%), 4-*Summary Statistics - Box Plots* (0%), 5-*Summary Statistics - No Graph* (0.8%) and 6-*Bivariate Data Analysis* (5.0%).
- This analysis looks at *Summary Statistics - Box Plots*.

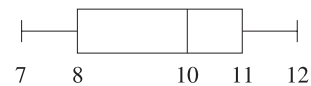
HSC ANALYSIS - What to expect and common pitfalls

- Box-plots* and in particular, the *comparison of parallel box-plots* receives an explicit mention in the new *Advanced* syllabus, making it surprising that this area is yet to be meaningfully examined in new syllabus exams.
- Box-plots* weren't examined in the 2021-23 Advanced exams and contributed a small part of a Bivariate Data question in the 2020 paper which is included in a later topic area.
- After minimal mark allocations between 2015-2018, *Box-plots* received a significant 5 mark question in the 2019 *Std2* exam, producing a mean mark of just 28%. Revision recommended here.
- In the last decade in *Std2* exams, students have been asked to compare parallel box plots in questions worth 4-6 marks no less than 3 separate times! This specific question style is a prime content area for a common Adv/Std2 common question and should feature in any revision.

Questions

1. Statistics, STD2 S1 2004 HSC 12 MC

This box-and-whisker plot represents a set of scores.



What is the interquartile range of this set of scores?

- A. 1
- B. 2
- C. 3
- D. 5

2. Statistics, STD2 S1 SM-Bank 2 MC

A dataset has the following five-number summary.

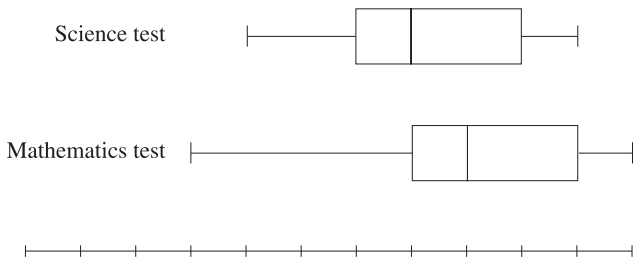
Minimum value	?
First quartile	10
Median	12
Third quartile	13
Maximum value	15

If the range of the dataset is 8, what is the minimum value of the dataset?

- A. 2
- B. 3
- C. 4
- D. 7

3. Statistics, STD2 S1 2008 HSC 10 MC

The marks for a Science test and a Mathematics test are presented in box-and-whisker plots.

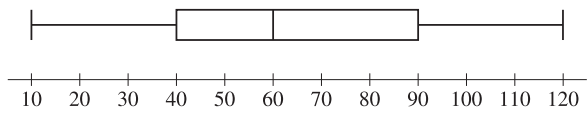


Which measure must be the same for both tests?

- A. Mean
- B. Range
- C. Median
- D. Interquartile range

4. Statistics, STD2 S1 2015 HSC 6 MC

The times, in minutes, that a large group of students spend on exercise per day are presented in the box-and-whisker plot.

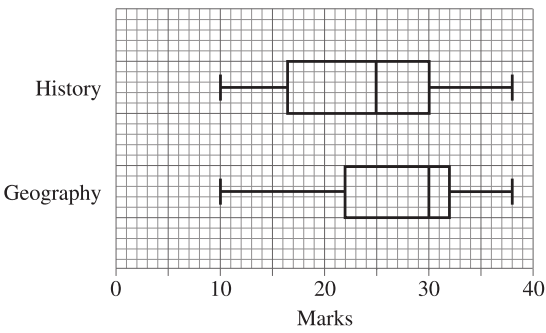


What percentage of these students spend between 40 minutes and 60 minutes per day on exercise?

- A. 17%
- B. 20%
- C. 25%
- D. 50%

5. Statistics, STD2 S1 2016 HSC 22 MC

The box-and-whisker plots show the results of a History test and a Geography test.



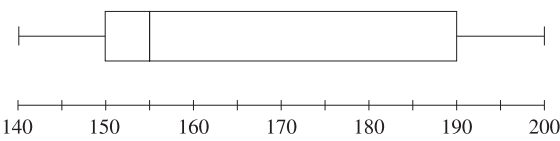
In History, 112 students completed the test. The number of students who scored above 30 marks was the same for the History test and the Geography test.

How many students completed the Geography test?

- A. 8
- B. 50
- C. 56
- D. 112

6. Statistics, STD2 S1 2011 HSC 7 MC

A set of data is displayed in this box-and-whisker plot.



Which of the following best describes this set of data?

- A. Symmetrical
- B. Positively skewed
- C. Negatively skewed
- D. Normally distributed

7. Statistics, STD2 S1 2016 HSC 19 MC

A soccer referee wrote down the number of goals scored in 9 different games during the season.

2, 3, 3, 3, 5, 5, 8, 9, ...

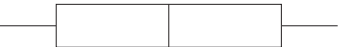
The last number has been omitted. The range of the data is 10.

What is the five-number summary for this data set?

- A. 2, 3, 5, 8.5, 12
- B. 2, 3, 5, 8.5, 10
- C. 2, 3, 5, 8, 12
- D. 2, 3, 5, 8, 10

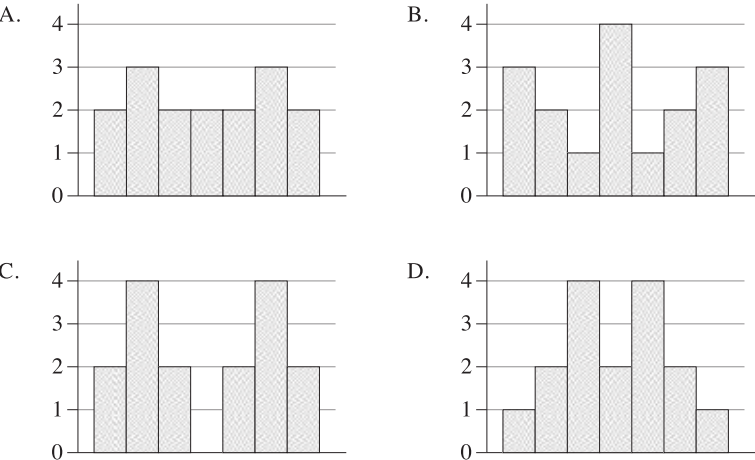
8. Statistics, 2ADV S2 2024 HSC 8 MC

Some data are used to create a box plot shown.



A histogram is created from the same set of data.

Which of these histograms is NOT possible for the given box plot?



9. Statistics, STD2 S1 2005 HSC 22 MC

Two groups of people were surveyed about their weekly wages. The results are shown in the box-and-whisker plots.



Which of the following statements is true for the people surveyed?

- A. The same percentage of people in each group earned more than \$325 per week.
- B. Approximately 75% of people under 21 years earned less than \$350 per week.
- C. Approximately 75% of people 21 years and older earned more than \$350 per week.
- D. Approximately 50% of people in each group earned between \$325 and \$350 per week.

10. Statistics, STD2 S1 SM-Bank 1

Write down the five-number summary for the dataset

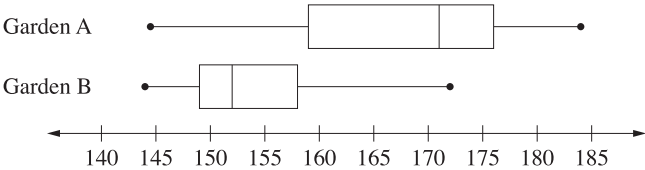
3, 7, 8, 11, 13, 18. (2 marks)

11. Statistics, 2ADV S2 2024 HSC 16

Flowers were planted in two gardens (Garden A and Garden B).

On a particular day, 25 flowers were randomly selected from each garden and their heights measured in millimetres.

The data are represented in parallel box-plots.



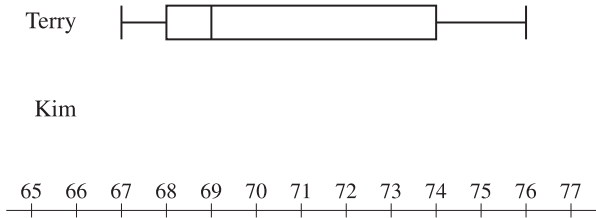
Compare the two datasets by examining the skewness of the distributions, and the measures of central tendency and spread. (3 marks)

12. Statistics, STD2 S1 2014 HSC 29c

Terry and Kim each sat twenty class tests. Terry's results on the tests are displayed in the box-and-whisker plot shown in part (i).

i. Kim's 5-number summary for the tests is 67, 69, 71, 73, 75.

Draw a box-and-whisker plot to display Kim's results below that of Terry's results. (1 mark)



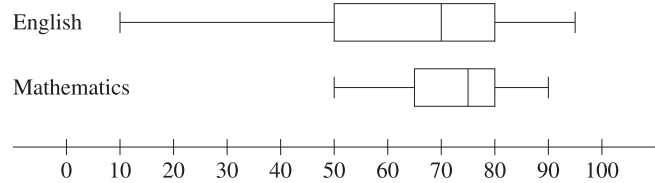
ii. What percentage of Terry's results were below 69? (1 mark)

iii. Terry claims that his results were better than Kim's. Is he correct?

Justify your answer by referring to the summary statistics and the skewness of the distributions. (4 marks)

13. Statistics, STD2 S1 2012 HSC 28d

The test results in English and Mathematics for a class were recorded and displayed in the box-and-whisker plots.

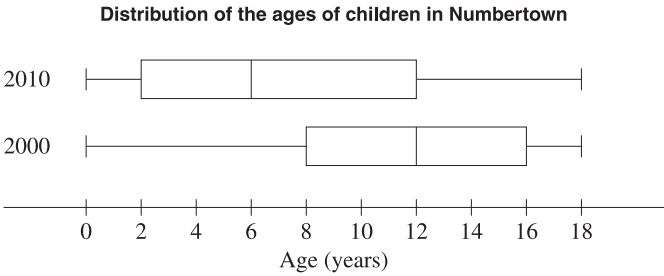


i. What is the interquartile range for English? (1 mark)

ii. Compare and contrast the two data sets by referring to the skewness of the distributions and the measures of location and spread. (3 marks)

14. Statistics, STD2 S1 2010 HSC 27b

The graphs show the distribution of the ages of children in Numbertown in 2000 and 2010.



i. In 2000 there were 1750 children aged 0–18 years.

How many children were aged 12–18 years in 2000? (1 mark)

ii. The number of children aged 12–18 years is the same in both 2000 and 2010.

How many children aged 0–18 years are there in 2010? (1 mark)

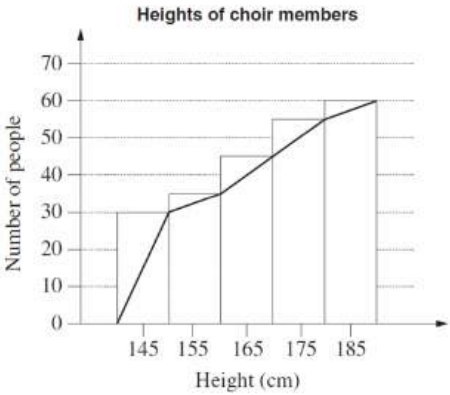
iii. Identify TWO changes in the distribution of ages between 2000 and 2010. In your answer, refer to measures of location or spread or the shape of the distributions. (2 marks)

iv. What would be ONE possible implication for government planning, as a consequence of this change in the distribution of ages? (1 mark)

15. Statistics, STD2 S1 2006 HSC 24c

The heights of the 60 members of a choir were recorded. These results were grouped and then displayed as a cumulative frequency histogram and polygon.

The shortest person in the choir is 140 cm and the tallest is 190 cm.



Draw an accurate box-and-whisker plot to represent the data. (3 marks)

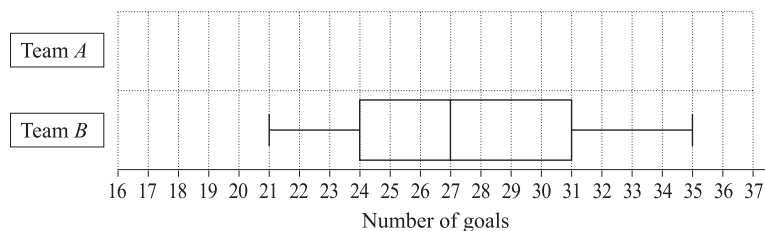
16. Statistics, STD2 S1 2019 HSC 39

Two netball teams, Team A and Team B, each played 15 games in a tournament. For each team, the number of goals scored in each game was recorded.

The frequency table shows the data for Team A.

<i>Number of goals</i>	<i>Frequency</i>
19	1
20	0
21	1
22	1
23	1
24	3
25	0
26	4
27	3
28	1

The data for Team B was analysed to create the box-plot shown.



Compare the distributions of the number of goals scored by the two teams. Support your answer with the construction of a box-plot for the data for Team A. **(5 marks)**

Worked Solutions

1. Statistics, STD2 S1 2004 HSC 12 MC

$$Q_1 = 8, Q_3 = 11$$

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 \\ &= 11 - 8 \\ &= 3 \end{aligned}$$

$$\Rightarrow C$$

2. Statistics, STD2 S1 SM-Bank 2 MC

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} \\ 8 &= 15 - \text{Min Value} \\ \therefore \text{Min} &= 15 - 8 \\ &= 7 \end{aligned}$$

$$\Rightarrow D$$

3. Statistics, STD2 S1 2008 HSC 10 MC

$$\begin{aligned} \text{IQR} &= \text{Upper Quartile} - \text{Lower Quartile} \\ \text{In both box plots, IQR} &= 3 \text{ intervals (against bottom scale)} \\ \Rightarrow D \end{aligned}$$

4. Statistics, STD2 S1 2015 HSC 6 MC

$$\begin{aligned} Q_1 &= 40, \text{ Median} = 60 \\ \therefore \% \text{ Students between 40 and 60} \\ &= 50\% - 25\% \\ &= 25\% \end{aligned}$$

$$\Rightarrow C$$

5. Statistics, STD2 S1 2016 HSC 22 MC

$$\begin{aligned} \text{In History} &\Rightarrow Q_3 = 30 \text{ marks} \\ \therefore \text{Scoring over 30} &= 25\% \times 112 = 28 \text{ students} \end{aligned}$$

$$\begin{aligned} \text{In Geography} &\Rightarrow \text{Median} = 30 \text{ marks} \\ \therefore \text{Students completing Geography} &= 2 \times 28 = 56 \text{ students} \\ \Rightarrow C \end{aligned}$$

6. Statistics, STD2 S1 2011 HSC 7 MC

Since the median (155) is closer to the lower quartile (150) and range low (140) than the upper quartile (190) and range high (200), it is positively skewed.
 $\Rightarrow B$

◆ Mean mark 47%.

7. Statistics, STD2 S1 2016 HSC 19 MC

Since range is 10 $\Rightarrow$ Last data point = 12
 $Q_1 = 3$
 $Q_3 = \frac{8 + 9}{2} = 8.5$
Median = 5
 $\Rightarrow A$

◆ Mean mark 46%.

8. Statistics, 2ADV S2 2024 HSC 8 MC

By inspection of box plot, $IQR = \frac{2}{3} \times \text{range}$
In options A, B and C (given 16 data points):
 Q_1 (4th data point) $\rightarrow$ 2nd column
 Q_3 (13th data point) $\rightarrow$ 6th column
Since $IQR = \frac{2}{3} \times \text{range} \Rightarrow$ histograms are possible.

In option D:
 $Q_1 \rightarrow$ 3rd column
 $Q_3 \rightarrow$ 5th column
Since $IQR = \frac{1}{3} \times \text{range} \Rightarrow$ not possible.
 $\Rightarrow D$

9. Statistics, STD2 S1 2005 HSC 22 MC

Option A: 50% of Under 21 group earned over \$325 and 75% of Over 21 group did. NOT TRUE.

Option B: 75% of Under 21 group earned below \$350 is TRUE.

Options C and D: can both be proven to be untrue using their median and quartile values.
 $\Rightarrow B$

10. Statistics, STD2 S1 SM-Bank 1

Minimum value: 3
First quartile: 7
Median: $\frac{11 + 8}{2} = 9.5$
Third quartile: 13
Maximum value: 18

11. Statistics, 2ADV S2 2024 HSC 16

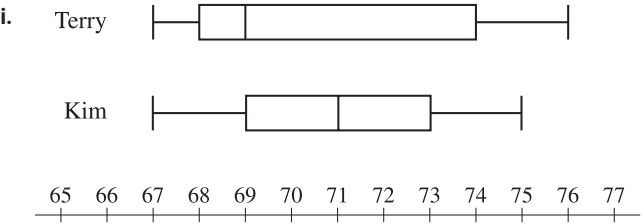
Skewness comparison:
 $\rightarrow$ Garden A is negatively skewed while Garden B is positively skewed.

Measures of central tendency comparison:
 $\rightarrow IQR(A) \approx 17$ is greater than $IQR(B) \approx 9$
 $\rightarrow$ This means the middle 50% of data points of B are more closely clustered around the median than A.

Spread comparison:
 $\rightarrow$ Range (A) ≈ 39 is less than range (B) ≈ 28
 $\rightarrow$ This means the spread of data points of A is wider than the spread of data points of B.

Overall, flowers from Garden B will tend to be higher than Garden A flowers as well as exhibiting a larger range of heights.

12. Statistics, STD2 S1 2014 HSC 29c



ii. 50%

◆ Mean mark 39%

iii. Terry’s results are more positively skewed than Kim’s and also have a higher limit high. However, Kim’s results are more consistent, showing a tighter IQR. They also have a significantly higher median than Terry’s and are evenly skewed.
∴ Kim’s results were better.

◆◆ Mean mark 29%
COMMENT: Examiners look favourably on using *language of location* in answers, particularly the areas they have specifically pointed students towards (skewness in this example).

13. Statistics, STD2 S1 2012 HSC 28d

i. $IQR_{(English)} = 80 - 50 = 30$

ii. Skewness

- English has greater negative skew
- Maths is more normally distributed

Location and Spread

- English has a range of 85, Maths has 40.
- English has larger IQR than Maths (30 vs 15)
- Maths’ median (75) is higher than English (70)
- Same upper quartile marks (80)
- English has highest and lowest individual mark

◆ Mean mark (ii) 35%
MARKER’S COMMENT: Markers are looking for students to use the correct language of location and spread such as *mean, median, interquartile range, standard deviation and skewness*.

14. Statistics, STD2 S1 2010 HSC 27b

i. Since the median = 12 years
⇒ 50% of children are aged 12–18 years
∴ Children aged 12–18 = 50% × 1750 = 875

◆ Mean mark (i) 45%

ii. Upper quartile (2010) = 12 years
Children in upper quartile = 875 (from part (i))
∴ Children aged 0–18 = 4 × 875 = 3500

◆◆ Mean mark (ii) 25%

- iii. Changes in distribution (include 2 of the following):
- the lower quartile age is lower in 2010
 - the median is lower in 2010
 - the upper quartile age is lower in 2010
 - the interquartile range is greater in 2010
 - 2010 is positively skewed while 2000 is negatively

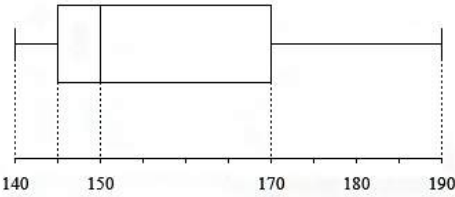
◆ Mean mark (iii) 35%
MARKER’S COMMENT: A number of students incorrectly identified “positive” skew as “negative” skew here.

- iv. Implication for government planning:
- Since the children are getting younger in 2010,
- Approve and build more childcare facilities
 - Build more school and public playgrounds

◆ Mean mark (iv) 46%
MARKER’S COMMENT: Answers should reflect the 1 mark allocation.

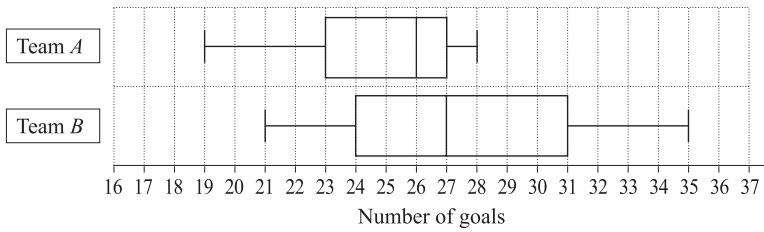
15. Statistics, STD2 S1 2006 HSC 24c

Low = 140
High = 190
Median = 150 (# People = 30)
 $Q_1 = 145$ (# People = 15)
 $Q_3 = 170$ (# People = 45)
Box and Whisker



16. Statistics, STD2 S1 2019 HSC 39

Team A: High = 28, Low = 19, $Q_1 = 23$, $Q_3 = 27$, Median = 26



Team A’s distribution is negatively skewed while Team B’s distribution is slightly positively skewed. The standard deviation of Team A’s distribution is smaller than Team B, as both its IQR and range is smaller. Team B is a more successful team at scoring goals as each value in its 5-point summary is higher than Team A’s equivalent value.

♦♦ Mean mark 28%.